

Jonathan E. Hunter, MChem. Ph.D.

Data Science | Chemical Informatics | Analytical Chemistry | Life Sciences

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Summary

Significant experience of cross-disciplinary and international research collaboration with award recognition and publications at the highest levels of academic impact — recent co-lead author publication in *Science*.

- **Delivering impactful insights through elegant solutions and life-long learning.**
- **Held prestigious positions in the USA, UK, and EU.**
- **Expert in liquid chromatography/mass spectrometry bioanalytical chemistry and metabolomics.**
- **Certifications and experience in Data Science and Chemical Informatics.**

Selected Skills

Category	Skills
Data Science & ML	Python (Pandas, Scikit-learn, Matplotlib), R, SQL, Git, Docker, AWS, Markdown, Bash, LaTeX
Metabolomics	XCMS, pyOpenMS
Scientific Research	Experimental Design, Microbiology, Sample Prep, LC/GC-MS Method Development
Communication & Analysis	Data Visualization, Manuscript Prep, Method Validation, Statistical Analysis
Leadership & Management	Research Project Management, Mentorship, Leadership

Professional Experience

Senior Scientist (Innovation), LGC, Cambridge, UK Feb. 2019 – Jan. 2023

- Led development of analytical methods for occupational drug testing in a regulated laboratory environment.
- Contract research for external sponsors, implementing novel technologies, and providing supervision, technical training, and mentorship to colleagues and students.
- Completed ISO17025 method validations for the determination of cannabinoids in strategically critical sample matrices and made key technical contributions to a large, commercially important contract research study on epicatechin metabolism.

These outcomes required complex problem-solving and close teamwork, conducted within rigorous scientific standards, meticulous record keeping, quality control, and efficiency considerations.

Postdoctoral Investigator, Woods Hole Oceanographic Institution, MA, USA Jul. 2016 – Jul. 2018

“Microscale Mechanistic Chemical and Physical Processes Contributing to Marine Organic Matter Degradation”

- Experimental design, cell culture, development of a novel nanoLC-MS methodology for high-sensitivity analysis, identification of unknown analytes, and quantification of target lipids.
- Collaborated with partner laboratories in Zurich and Calgary; presented findings at large international conferences and to philanthropic sponsors.
- Data analysis and visualisation using R, Excel, and MATLAB to handle large, multidimensional datasets, statistical interpretation, and broad chemical space.

Delivered publications at the highest levels of academic impact, including co-lead authorship in *Science* and lead authorship in *Environmental Science & Technology*.

Research Fellow, University of Southampton, UK

Jul. 2015 – Jun. 2016

“Characterization of Reactive Oxygen and Nitrogen Modified Lipids”

Visiting Student, AkzoNobel, NL

Jul. 2010 – Dec. 2010

“Synthesis of Metallocenes”

Developed process resulted in large-scale industrial production and was a commercial success.

Professional Service and Affiliations

- Peer reviewer for prestigious scientific journals, including *Nature Communications*.
- Member of the Royal Society of Chemistry (MRSC).

Selected Postgraduate Training

Certification	Date
Linux Commands & Shell Scripting Certification , IBM via EdX	Oct. 2024
Certified Cloud Practitioner Certification , AWS	Jun. 2024
Professional Data Scientist (Python) Certification , DataCamp	Jan. – May. 2024
Stepping up to Leadership Certification , Bespoke Training and Development	Jan. – May. 2022
*See DataCamp Portfolio for a full course list.	

Education

Doctor of Philosophy, Lipid Biochemistry, University of Southampton, UK

Aug. 2011 – Jul. 2015

“Phytoplankton Lipidomics: Lipid Dynamics in Response to Microalgal Stressors”

Master of Chemistry, University of Southampton, UK

Oct. 2007 – Jul. 2011

Honours and Awards

Award	Date
Spotlight Article – <i>Journal of Applied and Environmental Microbiology</i>	Jan. 2018
Outstanding Student Presentation Award – Aquatic Sciences Meeting, Granada, ES	Feb. 2015
Vice Chancellor’s Scholarship Award – University of Southampton	Jul. 2011
Bronze Award – International Genetically Engineered Machines (iGEM) Competition, MIT, MA, USA	Nov. 2009

Selected Publications

1. Behrendt L.†, Alcolombri U.†, **Hunter J. E.†**, et al. [“Microbial metabolic specificity controls pelagic lipid export efficiency”](#) *Science* (2024) & *bioRxiv* (2023). – A third-party perspective article on this work was published contemporaneously in *Science*.
2. **Hunter J. E.**, et al. [“Using high-sensitivity lipidomics to assess microscale heterogeneity in oceanic sinking particles and single phytoplankton cells”](#) *Environmental Science & Technology* (2021).
3. Laber C. P., **Hunter J. E.**, et al. [“Coccolithovirus facilitation of carbon export in the North Atlantic”](#) *Nature Microbiology* (2018).

4. **Hunter J. E., et al.** "Lipidomics of *Thalassiosira pseudonana* Under Phosphorus Stress Reveal Underlying Phospholipid Substitution Dynamics and Novel Diglycosylceramide Substitutes" *Applied and Environmental Microbiology* (2018).
5. **Hunter, J. E., et al.** "Targeted and Untargeted Lipidomics of *Emiliania huxleyi* Viral Infection and Life Cycle Phases Highlights Molecular Biomarkers of Infection, Susceptibility, and Ploidy" *Frontiers in Marine Science* (2015).

h-index = 5 with 270 citations as of Feb. 2025 ([Google Scholar](#))

†Authors contributed equally.

Selected Presentations

Presentation	Date
SSAS Department Meeting – LGC, Cambridge, UK "ISO17025 Method Development, Phytocannabinoids in Human Hair"	Jul. 2022
Ocean Sciences Meeting , Portland, OR, USA "Microscale Organic Matter Degradation Dynamics as determined by Single Particle Nano-Lipidomics"†	Feb. 2018
Simons Foundation – SCOPE Annual Meeting , New York City, NY, USA "ALOHA Nano-Lipidomics: Single Particle/Cell Lipidomics and Haptophyte Biomarkers"	Dec. 2017
Gordon and Betty Moore Foundation – Early Career Ocean Summit , Kona, HI, USA "Microscale Lipidomics of Organic Matter Degradation"	Jan. 2017
Society for Industrial and Applied Mathematics (Life Sciences Conference) , Boston, MA, USA "Chemical Signaling, Microbial Interactions, and the Ocean's Biological Carbon Pump"	Jul. 2016

† Presentation well recieved by an audience of >200 scientists.

Leadership and Mentorship

- Supervision and mentorship of undergraduate student projects.† **LGC & University of Southampton**
- Provided technical direction to junior scientists and laboratory technicians. **LGC & University of Southampton**

† Students awarded three 1st class Masters theses and one 2:1 Dissertation including the "Charnwood Molecular Prize for Best MChem Commercial Research Project".

Personal

- I am a husband and a father to two wonderful boys. We recently relocated to my hometown of Bristol to be closer to family.
- I have taken a career break to up-skill in Data Science and focus on my family, including a period as a full-time dad to my youngest son.
- In my free time, I enjoy mountain biking, running, chess, reading, cooking, and learning new skills.
- I consider myself to be a calm, quietly confident and friendly personality.

References available upon request.